

SCOPE OF THE DOCUMENT

This document is intended to support developers during the iC-MU Series design-in process. Based on our experience in typical applications the goal is to convey the information in a pragmatic way. Application specific values given in this document like - but not limited to - mechanical tolerances or tolerable analog errors are no definite values. They are meant as approximate guide values. Such values need to be verified and validated in the light of the specific application, the specific performance criteria and overall required operating conditions.



This document is only applicable for iC-MU Series GUI revision B1 or later.

GUI revision	Applicable iC-MU AN3 revision
B1 or later	revision D1 or later
Ax	revision Cx (on request)

Table 1: Applicable iC-MU AN3 revision depending on the iC-MU Series GUI revision.

TOOLS DOCUMENTATION FOR CALIBRATION AND PROGRAMMING

1. iC-MU product page (www.ichaus.com/mu)
2. iC-MU150 product page (www.ichaus.com/mu150)
3. iC-MU200 product page (www.ichaus.com/mu200)
4. iC-Haus USB PC Adapter (www.ichaus.com/tools)

Please see the product page download section for iC-MU Series GUI, DLL, MCU driver and further documentation. All described procedures and settings in this application note apply to the iC-MU Series chip family.



The iC-MU Series GUI is based on the iC-MU Series DLL. The DLL can be used to implement all iC-MU Series specific configuration and calibration functionality in own software solutions.

Please check Table 2 below for USB BiSS/SPI adapter compatibility with the iC-MU Series GUI software and DLL for calibration.

USB PC adapter	BiSS calibration	SPI calibration
MB4U (discontinued)	Yes	No
MB5U	Yes	No
MB3U-I2C	No	Yes

Table 2: USB PC adapter compatibility matrix for iC-MU Series calibration

MECHANICAL TOLERANCES

Understanding the allowable mechanical tolerances of an iC-MU Series based system is required for a successful calibration and reliable product implementation.

The tolerances given below are **static** tolerances. These tolerances refer to the acceptable mechanical misalignment that can be accepted during the initial assembly of an iC-MU Series based encoder system. Analog signal errors or Nonius mismatch caused by these static mechanical mounting tolerances can be compensated by the analog and Nonius calibration.

Mechanical play that occurs during the operation of an iC-MU Series based encoder system (dynamic mechanical tolerances) has a negative impact on the system performance in terms of position accuracy and Nonius reserve.

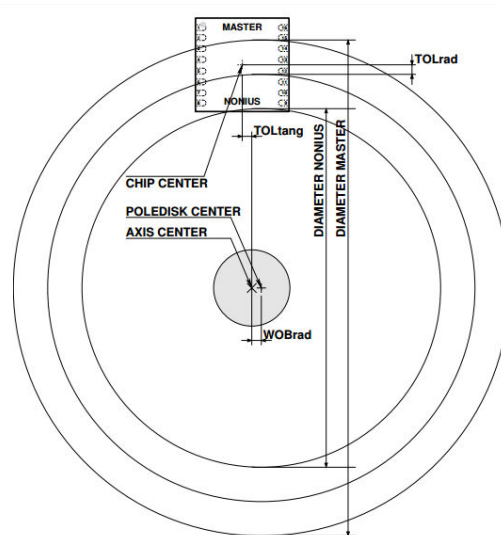


Figure 1: Definition of radial, tangential and eccentricity tolerances

iC-MU Series datasheet Item No.	Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
301	TOLrad	Permissible Radial Displacement				0.5	mm
302	TOLtang	Permissible Tangential Displacement				0.5	mm
303	WOBrad	Permissible eccentricity of the magnetic pole wheel	MPC = 0x4 MPC = 0x5, 0x6			0.06 0.1	mm
	P2	iC-MU package surface to magnetic coating surface distance	all package variants	0.2	0.4	0.6	mm
	P2	iC-MU150 package surface to magnetic coating surface distance	all package variants	0.2	0.5	0.7	mm
	P2	iC-MU200 package surface to magnetic coating surface distance	all package variants	0.3	0.6	0.8	mm

Table 3: Permissible static mechanical alignment tolerances for initial assembly



The P2 (package surface to magnetic coating surface distance) values given above are recommendations only. The suitability of these values needs to be evaluated and verified in the final application on a case to case basis.

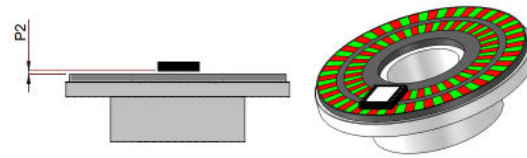


Figure 2: Air gap iC-MU Series to magnetic pole wheel

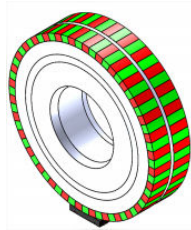


Figure 3: Radial magnetic pole wheel example

Please refer to iC-MU Series magnetic pole wheel specifications for positioning information of iC-MU Series relative to the magnetic pole wheel.

SETUP

The following assumes that the BiSS interface with adaptive timeout is used for calibration. The basic procedure is also valid for calibration via the SPI interface, SPI specific settings have to be considered accordingly.

1. Mount the iC-MU Series magnetic pole wheel and MU1M module according to the mechanical specifications in chapter "MECHANICAL TOLERANCES".
2. Connect the ribbon cable from the MU1M to the MU1D board connector J1.
3. Connect the MB5U adapter to the MU1D board connector J2.
4. Ensure that JP1, JP5 and JP6 are closed. It is recommended to power the MU1D board via an external stabilized and filtered power supply. Ensure that JP2 and JP3 are open in that case. If no external power supply is available the board can also be powered via connector J2 using the MB5U adapter. In this case ensure that JP2 is closed.

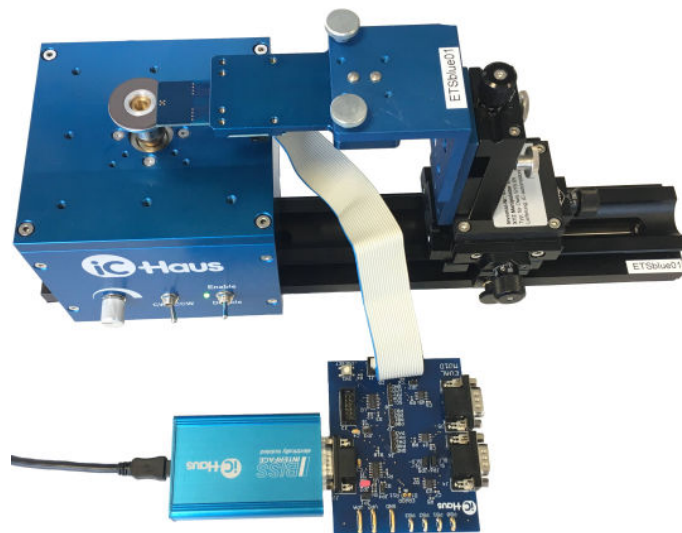


Figure 4: Example iC-MU Series test setup allowing X-, Y- and Z-Axis adjustments

5. Launch the iC-MU Series GUI. (Always ensure that the latest software version is used. It can be found at www.ichaus.com/software).
6. Connect to the iC-MU Series chip by pressing the <Disconnected> button. Verify that the status changes to <Connected>.



In delivery state MU1M and MU6M are configured to SPI interface. Please refer to the respective evaluation kit specification on how to connect to iC-MU Series via the BiSS interface.

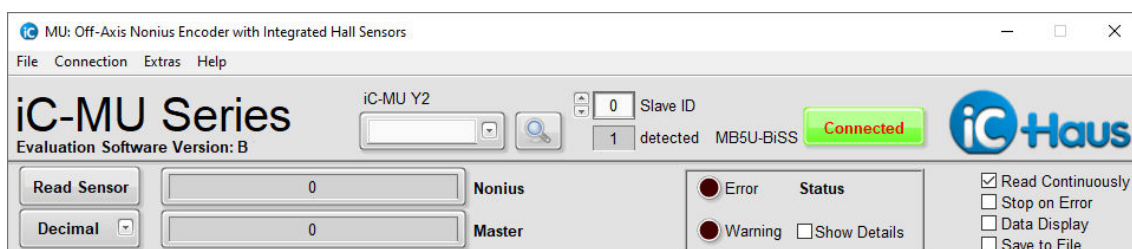


Figure 5: iC-MU Series GUI - Successful connection to the iC-MU Series chip

7. When using the MB5U adapter calibration can only be carried out in BiSS mode. For SPI calibration please use the MB3U-I2C adapter. Please verify that <Port A Configuration> under tab <Interface> is configured to the interface that is to be used for calibration.

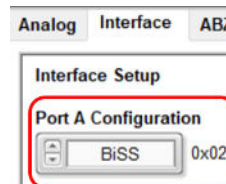


Figure 6: iC-MU Series GUI, Port A configured to BiSS

8. Ensure that <Amplitude Control> is enabled under tab <Analog>.

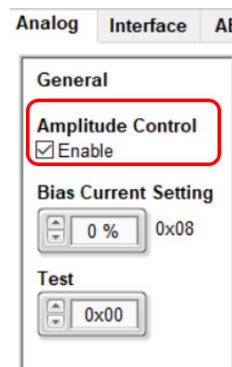


Figure 7: iC-MU Series GUI, Amplitude control enabled

9. Under the <Nonius> tab, select the following:
- Hall Sensor Arrangement: Select <Rotative> for use with axial magnetic pole wheels (e.g. MU2S, MU18S...) or <Linear> for use with radial magnetic pole wheels or linear magnetic tapes (e.g. MU7S, MU1L, MU2L...).
 - Set the Master Period Count according to the magnetic pole wheel used (e.g. 32 master track magnetic pole pairs for MU2S / MU18S).
 - Filter Features: Select 39dB = 0x04. This is the recommended starting point for conducting the calibration procedure.

It is permissible to use a different filter setting for the calibration procedure. Please note that...



- higher filter settings (e.g. 45dB or 51dB) can reduce noise on the Nonius curve but might prolong the settling time of the analog calibration algorithm.
- lower filter settings (e.g. 27dB or 21dB) can increase noise on the Nonius curve but might shorten the settling time of the analog calibration algorithm.

iC-MU AN3

ROTARY CALIBRATION AND PROGRAMMING



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e.g.
MU2S 30-32N
MU18S 30-32N
MU36S 35-32N

Converter Nonius Calculation
Hall Sensor Arrangement

☒ Rotative (axial)
☐ Linear/Rotative (radial)

e.g.
MU7S 25-32N
MU26S 52-64N
MU1L 41-16N

Master Period Count

32

0x05

Filter Features

51 dB

0x06

**Automatic Reset
triggered by AM_MIN**

☐ Enable

Nonius Period Verification

☐ Disable

Figure 8: iC-MU Series GUI, <Nonius> tab settings

10. Under tab <CalibSetup> select <Singleturn Calibration>.

Analog Interface ABZ / UVW Nonius Multiturn Nonius Calibration Error / Warning / Status Misc. I2C CalibSetup Hex Editor

Calibration Mode

☒ Singleturn Calibration
☐ 3-Track Nonius Calibration

Calibration Setup

Settings

3-Track Nonius Calibration Wizard

Calibrate

Figure 9: iC-MU Series GUI, <CalibSetup> tab settings

CALIBRATION

11. Calibrate the iC-MU BIAS current as described in the iC-MU Series specification.



The BIAS current calibration is used to equalize chip to chip variations mainly due to the chip manufacturing tolerances. It is prerequisite for adherence to the given electrical characteristics in the datasheet and also instrumental in the determination of the chip timings. The bias current only needs to be calibrated once, recalibration during the lifetime of the device is not necessary.

12. The interface settings for the calibration data acquisition can be adjusted under tab <Connection> → <Calibration Setup>, see Figure 10. Alternatively, the <Settings> buttons in the <Analog> and <Nonius> tabs can be used. Please also see chapter DETERMINATION OF THE CALIBRATION PARAMETERS for further information on SPI and BiSS calibration settings.

Calibration Data Acquisition Settings

<SPI Frequency>	This is the SPI clock frequency for SPI based calibration. The resulting frame repetition rate is calculated from the SPI clock frequency and the payload length. The value displayed corresponds to the calibration data sampling rate. See chapter DETERMINATION OF THE CALIBRATION PARAMETERS for further information.
<BiSS Frequency>	The BiSS clock frequency must be selected so that the desired BiSS frame rate can be achieved. In case the BiSS frequency is too low to support the selected BiSS frame rate an error message is generated by the GUI. A setting of 3.3 MHz is typically appropriate.
<BiSS Frame Repetition Rate>	Corresponds to the BiSS calibration data sampling rate, please see chapter DETERMINATION OF THE CALIBRATION PARAMETERS for further information.
<Acquisition Time>	Determines the duration of the calibration data acquisition in seconds. For rotary systems the time should be selected so that multiple full revolutions of the magnetic pole wheel are measured. In linear measurement systems it should be chosen so that the linear measurement range is scanned multiple times. The higher the number of recorded revolutions or linear measurement ranges, the better the data averaging.



To avoid USB data loss during calibration data acquisition with the iC-Haus USB PC adapter...

- do not connect any other USB devices to the same USB bus as the adapter.
- close any unnecessary processes on the PC.

Calibration Quality Check Settings

The Calibration Quality Check is an optional feature that can be enabled via the corresponding checkbox. When enabled it checks whether the selected calibration criteria are met to ensure consistent calibration quality.

Minimum desired rotative mechanical revolutions / rotative mechanical degrees / linear absolute measurement distance

This criteria determines the minimum number of mechanical revolutions or absolute linear measurement distance to be recorded for calibration. Note that the linear distance can't be entered directly, 1 rev. or 360°m equals the full absolute linear measurement distance.

Minimum desired number of interpolation samples per electrical master period

This criteria determines the minimum number of samples per electrical master period to be recorded for calibration. Default value is 128, please see chapter DETERMINATION OF THE CALIBRATION PARAMETERS for further information.

Calculated Speed Range

The calculated speed range min/max values indicate in which rotative or linear speed range the selected quality criteria are met. The values are calculated based on the selected frame rate, data acquisition time, desired number of revolutions / linear measurement distance, minimum number of samples per master period, parameter MPC and the magnetic pole pitch.

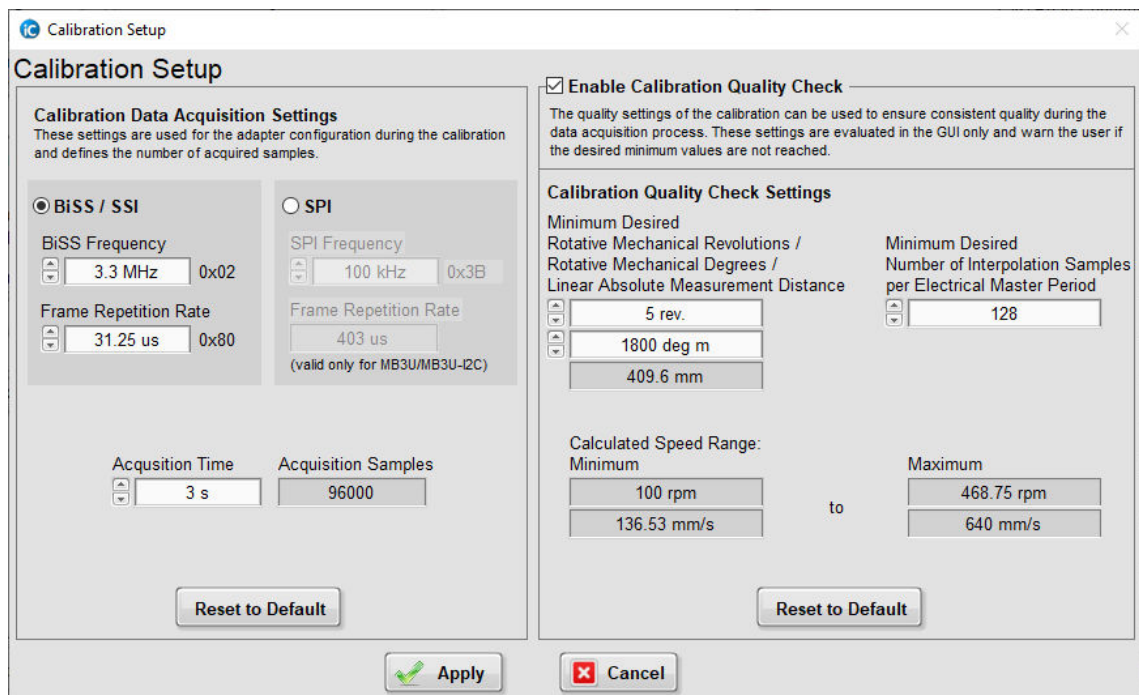


Figure 10: iC-MU Series GUI, Calibration Setup

13. The analog calibration should be started with all analog signal conditioning parameters at their default value. The <Reset> button on the <Analog> tab can be used to reset the parameters to their default values.

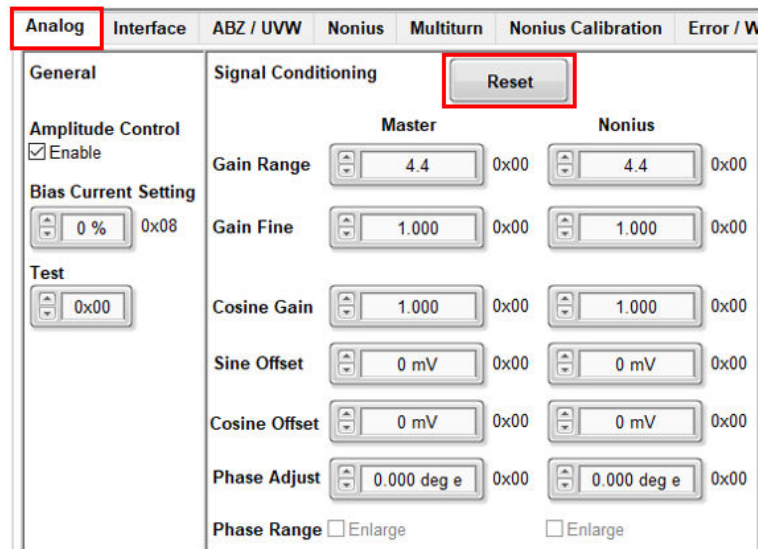
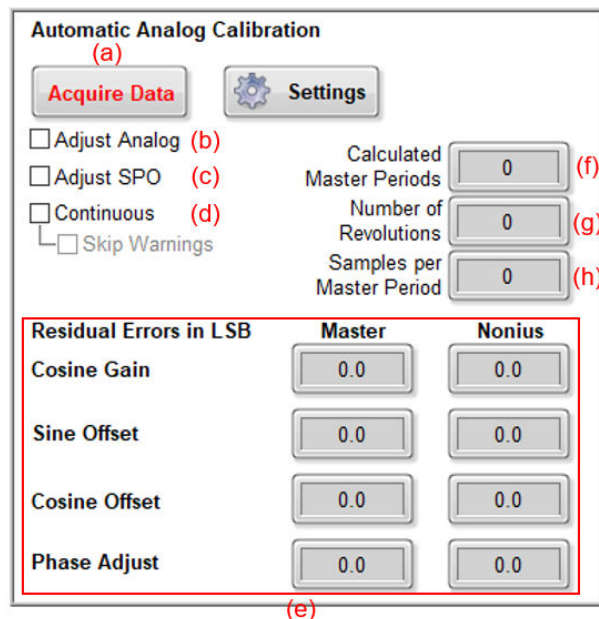


Figure 11: iC-MU Series GUI, analog calibration - setting default values



For rotary radial or linear encoder systems it may be reasonable to start the analog calibration process with <Phase Adjust> values other than the default values. This is the case if a certain phase correction value is anticipated due to an already known pole pitch mismatch of the magnetic pole wheel / tape. For examples please check the respective radial / linear default .cfg files included in the GUI installation <data> folder.

14. The "Automatic Analog Calibration" window is situated on the right hand side of the <Analog> tab. The individual controls and displays are explained below:
 - (a) <Acquire Data> - This button starts the calibration routine.
 - (b) <Adjust Analog> - When selected the analog calibration is carried out and the signal conditioning parameters (GX_x, VOSS_x, VOSC_x, PH_x, PHR_x) are adjusted accordingly. When the checkbox is deselected the current analog errors are measured only. The signal conditioning parameters remain untouched in that case.
 - (c) <Adjust SPO> - When selected the Nonius calibration is carried out and the SPO parameters are adjusted accordingly. When the checkbox is deselected the current Nonius curve is measured only. The SPO parameters remain untouched in that case.
 - (d) <Continuous> - When selected the calibration or measurement cycle is repeated until manually stopped. This can be especially useful to continuously observe analog error changes or changes in the Nonius curve when <Adjust Analog> and <Adjust SPO> are both deselected, thus sole measurements are conducted.
 - (e) "Residual Errors in LSB" - The residual errors after the calibration routine has been carried out. When <Adjust Analog> is deactivated the measured uncompensated analog errors are displayed.
 - (f) "Calculated Master Periods" - The number of magnetic master periods determined by the algorithm. The value needs to match the number of master periods selected by parameter <MPC>.
 - (g) "Number Of Revolutions" - Displays how many full mechanical pole wheel revolutions have been recorded during calibration data acquisition. This should always be multiple rotations for rotary systems.
 - (h) "Samples per Master Period" - Displays the minimum number of samples per magnetic master period that have been recorded during the calibration data acquisition. Please see chapter DETERMINATION OF THE CALIBRATION PARAMETERS for further information.



The screenshot shows the 'Automatic Analog Calibration' window. At the top, there are two buttons: 'Acquire Data' (labeled (a)) and 'Settings' (labeled (b)). Below these are three checkboxes: 'Adjust Analog' (labeled (b)), 'Adjust SPO' (labeled (c)), and 'Continuous' (labeled (d)). There is also an unchecked checkbox for 'Skip Warnings'. To the right of these checkboxes are three input fields: 'Calculated Master Periods' (labeled (f)) with value 0, 'Number of Revolutions' (labeled (g)) with value 0, and 'Samples per Master Period' (labeled (h)) with value 0. At the bottom, there is a section titled 'Residual Errors in LSB' (labeled (e)) which contains a table with two columns: 'Master' and 'Nonius'. The rows in the table are 'Cosine Gain', 'Sine Offset', 'Cosine Offset', and 'Phase Adjust', each with input fields showing 0.0.

Figure 12: iC-MU Series GUI, Automatic Analog Calibration

15. Rotate the iC-MU Series magnetic pole wheel (please see chapter DETERMINATION OF THE CALIBRATION PARAMETERS regarding permissible rotation speed or use the calibration quality check feature). Ensure that both the <Adjust Analog> and <Adjust SPO> checkboxes are selected. Press <Acquire Data> to start the calibration routine.

i If desired it is also possible to conduct the analog and Nonius calibration individually by only selecting the <Adjust Analog> or <Adjust SPO> checkbox. Since both calibrations are necessary for a properly functioning system, it is more convenient to perform both in one pass.

! If an external Multiturn is configured please deactivate the cyclic multiturn verification during the analog and Nonius calibration (CHK_MT = 0x00). This prevents that additional noise is introduced by the multiturn data transmission during calibration.

Both the analog and Nonius calibration are now carried out automatically. Calibration data is acquired and the analog compensation values and <SPO> parameters are calculated and adjusted. Finally another measurement is automatically taken to determine the residual analog errors and display the compensated Nonius curve on the <Nonius Calibration> tab. Please ensure that the magnetic pole wheel is rotating until the calibration routine is completely finished.

The rotary motion profile is not critical, but it must be ensured that the requirements in chapter DETERMINATION OF THE CALIBRATION PARAMETERS are met. A change in the direction of rotation or a change in rotational speed are well tolerated by the calibration algorithm. Stops are also tolerated but should be avoided as they reduce the number of usable data points.

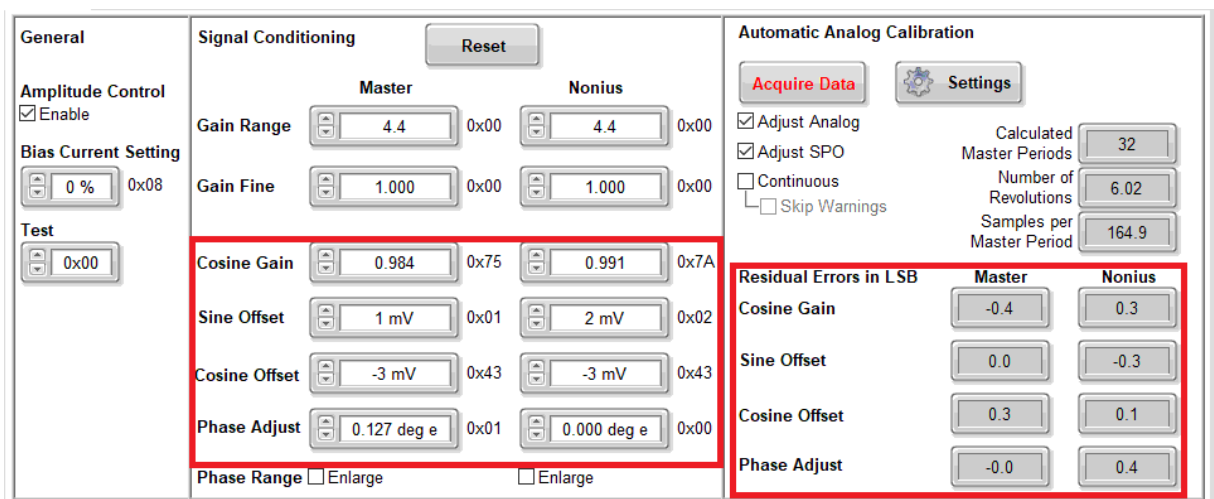
i If a full mechanical rotation is not possible, only a section of a rotary pole wheel can be scanned for calibration. It is important to scan the whole section which is to be used later in the application.

16. Review the analog calibration result. The "Residual Errors in LSB" are shown on the right hand side of the window (see Fig. 13). Ideally these values should be as close to zero as possible, indicating that the analog

calibration algorithm determined suitable signal conditioning values. For residual error values significantly greater (e.g. ± 2 LSB or more) the calibration routine may be repeated multiple times to further minimize the residual error.

If the residual error keeps changing and never settles further troubleshooting of the system should be carried out. Some common issues are for example:

- Poor mechanical setup of the system causing excessive mechanical play during calibration.
- Bad quality or damaged magnetic pole wheel.
- Inapt calibration data acquisition settings resulting in an insufficient number of data points per magnetic period.
- Excessive cogging torque when rotating the magnetic pole wheel.



Signal Conditioning		Automatic Analog Calibration	
Master	Nonius	Settings	Calculated
Gain Range: 4.4	Gain Range: 4.4	☒ Adjust Analog	Master Periods: 32
Gain Fine: 1.000	Gain Fine: 1.000	☒ Adjust SPO	Number of Revolutions: 6.02
		☐ Continuous	Samples per Master Period: 164.9
		☐ Skip Warnings	
Residual Errors in LSB			
Cosine Gain: 0.984	Cosine Gain: 0.991		
Sine Offset: 1 mV	Sine Offset: 2 mV		
Cosine Offset: -3 mV	Cosine Offset: -3 mV		
Phase Adjust: 0.127 deg e	Phase Adjust: 0.000 deg e		

Figure 13: iC-MU Series GUI, Analog Calibration - Signal conditioning values and residual errors in LSB after successful analog calibration

On the left hand "Signal Conditioning" side of the window the analog signal conditioning parameters are set to the values determined by the calibration algorithm. The closer these values are to their default values the better the alignment of the iC-MU Series chip to the magnetic pole wheel is. Well aligned systems typically yield the following results:

- Sine and cosine master track offset values are within ± 10 mV.
- Sine and cosine Nonius track offset values are within ± 20 mV.
- Both phase adjust values are within $\pm 2^\circ$. For systems with an expected systematic phase error (e.g. radial rotary scanning / linear scanning) the values should be $\pm 2^\circ$ relative to the expected value.

The analog calibration results shown in Figure 13 are considered very good.



The offset and phase signal conditioning limits above are only to be seen as a recommendation. The internal circuitry can compensate for even larger errors. As long as the control limits are not reached the system might still yield acceptable overall performance with a less accurate initial mechanical alignment. This needs to be evaluated and verified in the final application on a case to case basis. Nevertheless it is always advisable to mechanically adjust the system as accurately as possible in the respective application.

17. If the sine/cosine offset or phase adjust signal conditioning values are significantly above the recommended values consider the following:

- **Sine/Cosine offset:** Adjust the distance of the iC-MU Series package to the magnetic pole wheel surface (air gap). Typical values are given in Table 3 in chapter "MECHANICAL TOLERANCES". After adjustment repeat the analog calibration starting at Step 13.
- **Phase Adjust:** Adjust the mechanical position (radial, tangential, or both) of the iC-MU Series chip to the magnetic pole wheel and redo the analog calibration starting at Step 13.

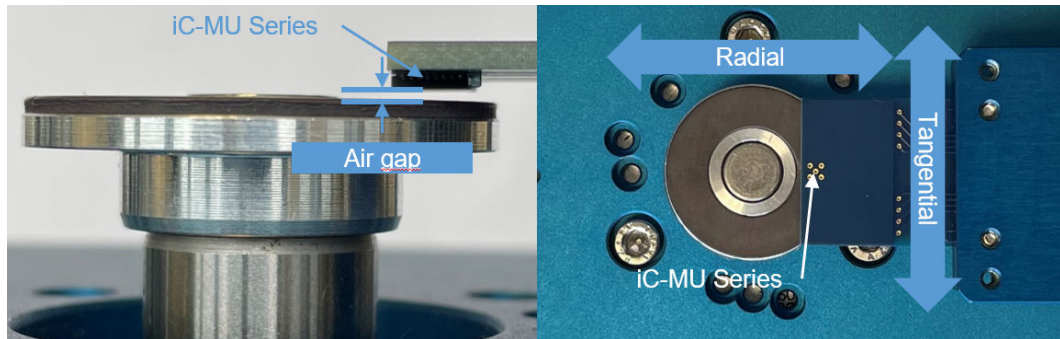


Figure 14: iC-MU Series - Mechanical adjustment

18. After the analog calibration has been successfully performed, the result of the Nonius calibration (Adjust SPO) needs to be checked. Change to the <Nonius Calibration> tab.

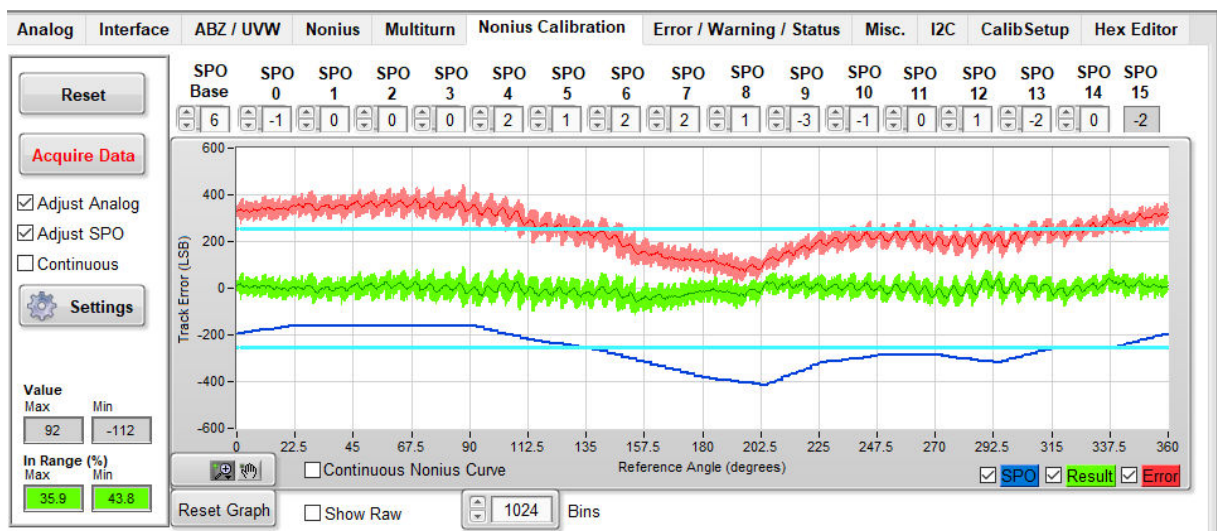


Figure 15: iC-MU Series GUI, Nonius calibration result

If selected three curves are shown:

- **Red curve (Error):** This is the uncorrected Nonius error curve
- **Blue curve (SPO):** This is the calculated correction curve
- **Green curve (Result):** This is the corrected Nonius curve.

The Nonius calibration is successful when the resulting green curve is centered between the limits marked by the bright blue lines with enough margin to either side. Keeping both the "In Range Max" and "In Range Min" values below 60% is recommended. This is equal to a Nonius margin of at least 40% to either side. Figure 16 below shows the definition of the nonius margin.

upper margin = 100% - "In Range Max"

lower margin = 100% - "In Range Min"

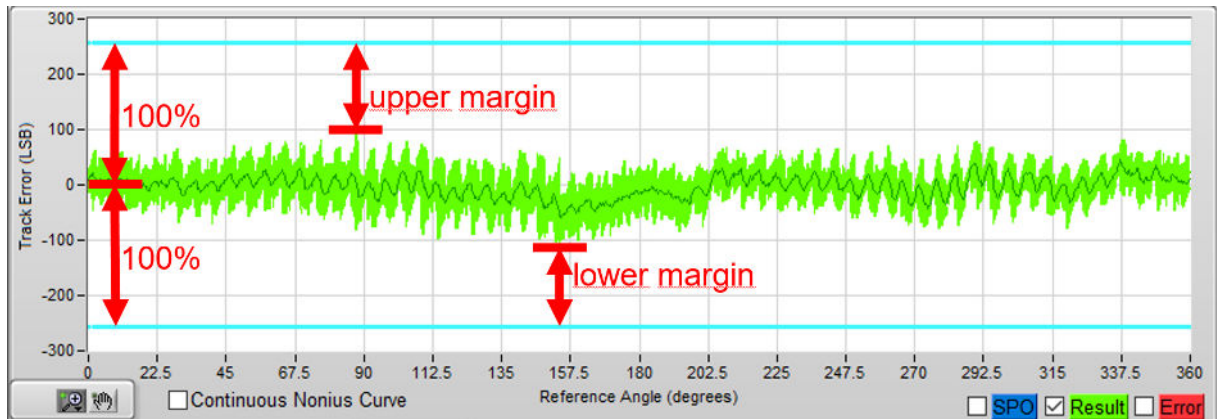


Figure 16: Calculation of the Nonius phase margin

The Nonius curve is affected by mechanical movement of the iC-MU Series relative to the magnetic pole wheel after the calibration has been conducted. Always ensure that the Nonius curve stays within the required limits under all operating conditions (especially temperature range) of the encoder system.

This can be checked by deselecting the <Adjust Analog> and <Adjust SPO> checkboxes, rotating the magnetic pole wheel and pressing the <Acquire Data> button. This performs a measurement only and allows to monitor the Nonius curve without adjustments to the signal conditioning or SPO parameters. The checkbox <Continuous> may be selected to loop the measurements until manually stopped.

When <Continuous Nonius Curve> is selected, the Nonius curve is displayed continuously over several revolutions. This feature can be useful to monitor a mechanical movement of the iC-MU Series relative to the magnetic pole wheel that happens over multiple rotations.

See Figure 17 for an example of the effect of a mechanical wobble over multiple rotations on the continuous Nonius curve.

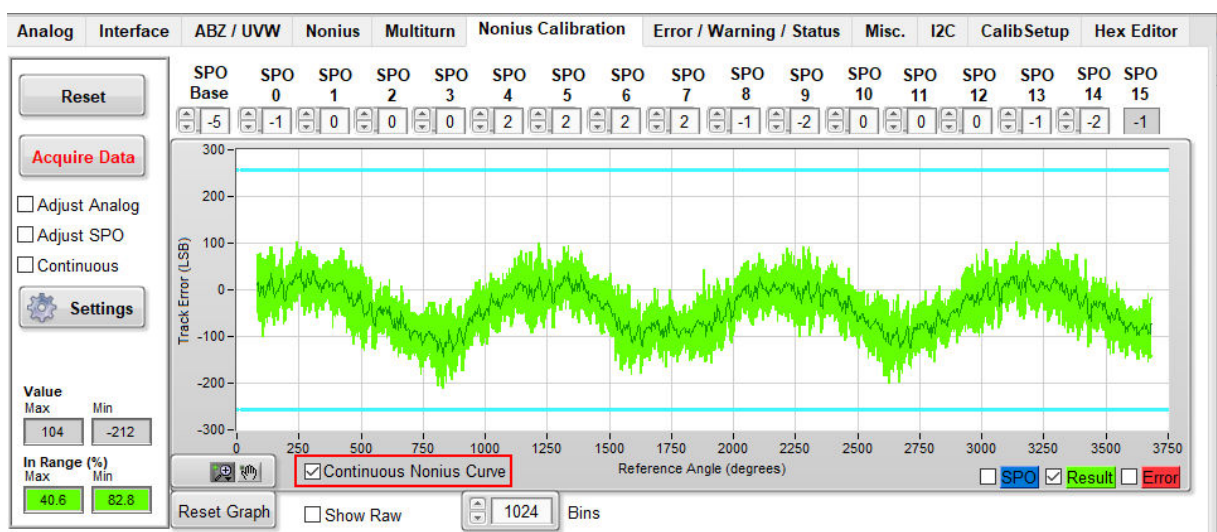


Figure 17: Example of continuous Nonius curve with mechanical wobble

If the Nonius margin is insufficient, further troubleshooting of the system is necessary. Some common issues are:

- Poor chip to magnetic pole wheel alignment (radial, tangential or axial).
- Excessive mechanical tolerances of the encoder system.
- Poor quality of the magnetic pole wheel.

- Poor or missing analog calibration.
 - Incorrect chip configuration.
19. After successful Nonius calibration select the <Error/Warning/Status> tab (Figure 18) and perform the following:
- (a) Under Status Register Configuration select <Accumulated>. This causes all status flags to be latched. The status flags are only reset if the status register is read and the underlying error condition is no longer present.
 - (b) Select the <Show Details> checkbox. This will bring up the Status Information Window (see Figure 19).
 - (c) Rotate the magnetic pole wheel and press the <Read Status> button in the Status Information Window after at least one full rotation. No errors should be present.

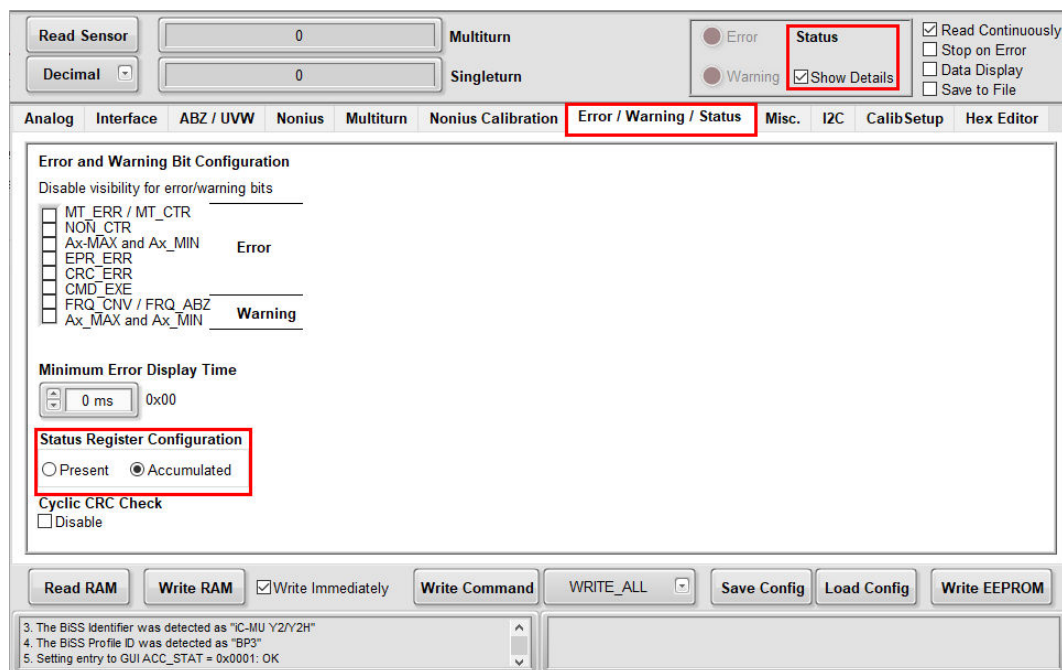


Figure 18: iC-MU GUI, <Error/Warning/Status> tab settings

Below is a summary of some common status flags that might be set at this point. Please always consult the respective iC-MU Series datasheet for in depth information on the status register:

- **Signal Error Clipping or Poor Level (AN_MAX, AM_MAX, AN_MIN, AM_MIN):** This indicates that the iC-MU-Series automatic amplitude control is close to the control limits. Ensure that the air gap is adjusted according to Table 3.
- **Invalid checksum internal RAM (CRC_ERR):** The CRC_ERR is set if the cyclic CRC check is configured (NCHK_CRC=0) and changes to the RAM parameter settings are made without recalculating the checksum (command CRC_CALC) or writing to the EEPROM (command WRITE_ALL). This status flag may be safely ignored if iC-MU Series parameters are still adjusted during the setup and calibration procedure.
- **Period counter consistency error (NON_CTR):** This status flag is active if the computation of the absolute position fails at certain points of the magnetic code wheel. The absolute position may be incorrect. Check the the Nonius phase margin.

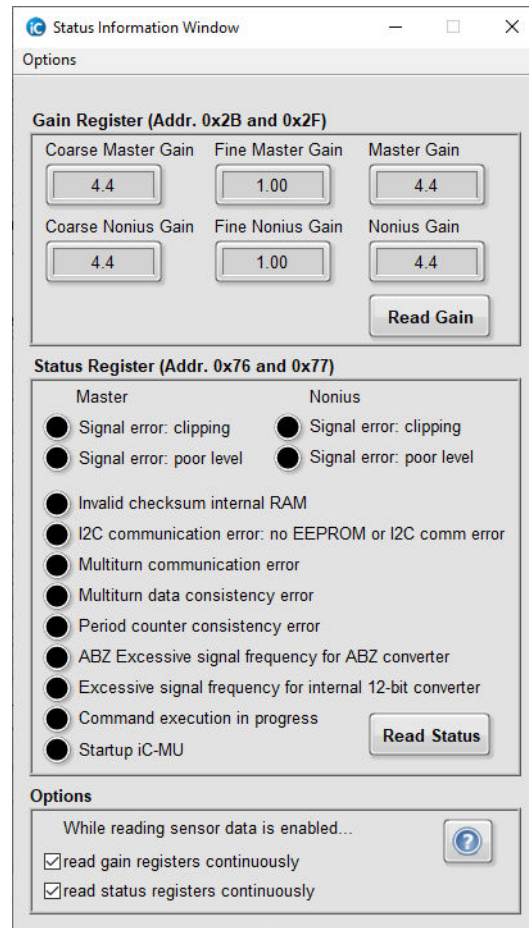


Figure 19: iC-MU Series GUI, Status Information Window

20. After successful analog and Nonius Calibration and with no active Error/Warning flags the calibration is complete. Press <Write EEPROM> at the bottom of the GUI screen to store the settings in the connected EEPROM.



Figure 20: iC-MU Series GUI, Write EEPROM button

21. The calibration is now complete.



It is recommended to always check the quality of the calibration with suitable external measurement tools.

LINEAR CALIBRATION

The calibration procedure for linear measurement applications is mostly identical to the calibration of rotative applications. Differences are highlighted below.

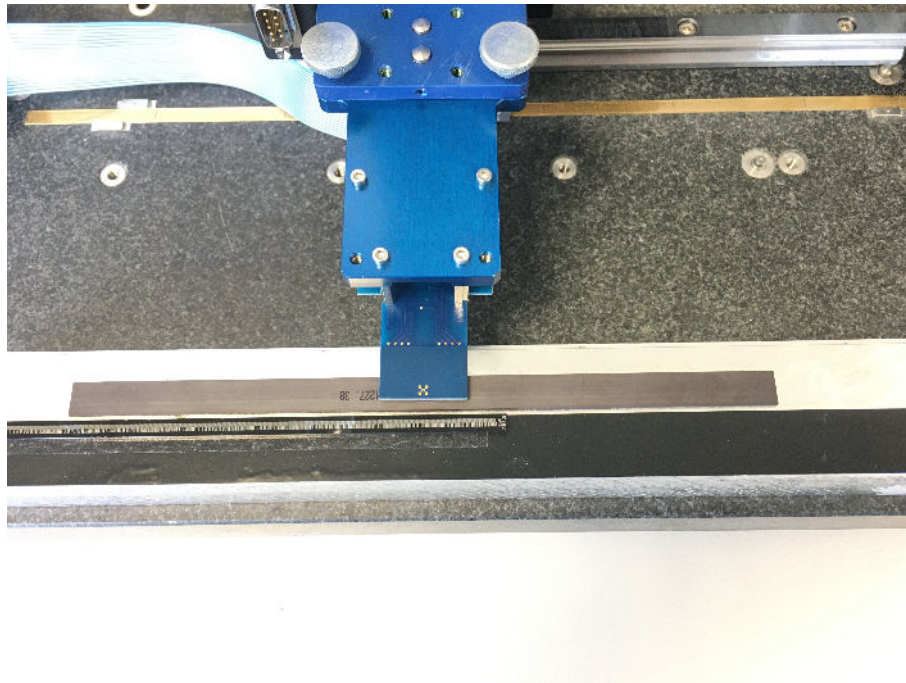


Figure 21: Linear measurement setup example

1. Ensure that <Linear/Rotative (Radial)> is selected under Hall Sensor Arrangement in Step 9. All other settings in Step 9 and 10 can stay the same as for the rotary calibration.
2. Choose the <Calibration Time> in Step 12 so that the linear measurement range is scanned multiple times. Please note that the linear motion needs to continue until the calibration routine is completely finished.



It is important to scan the whole magnetic tape section which is to be used later in the application.

3. Start the linear movement and conduct the calibration routine as per Step 15. As for the rotary motion profile the linear motion profile is also not critical.
4. Evaluate the analog and Nonius calibration and finalize the calibration procedure as described in Step 16 and following. Please note that with linear measurement systems an excessive phase error (step 17) can be caused by a rotation of iC-MU Series to the magnetic tape about its own axis (see Figure 22).



Figure 22: Linear measurement setup - Rotation of iC-MU Series about its own axis

ROTARY CALIBRATION PROCEDURE FLOWCHART

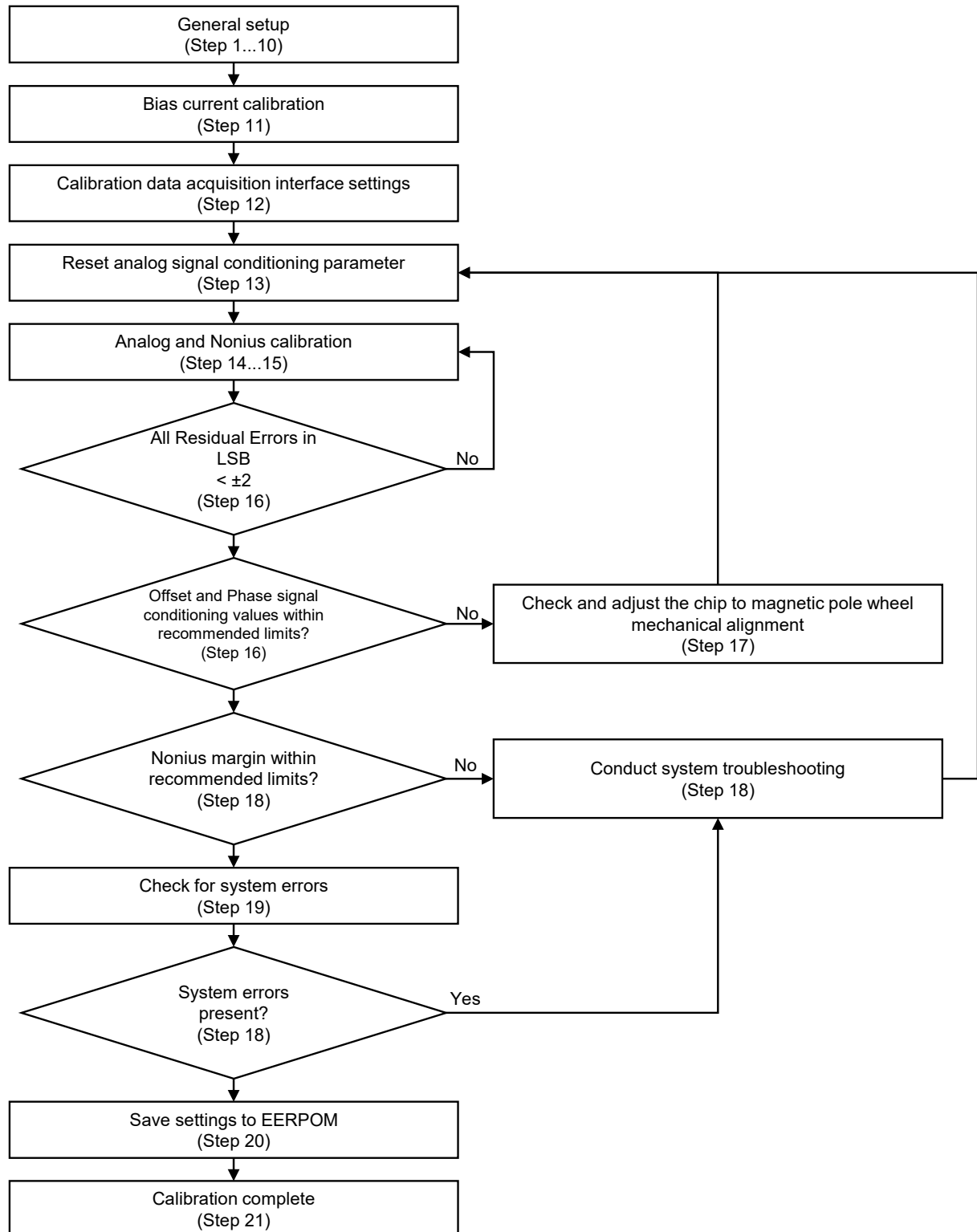


Figure 23: Rotary Calibration Procedure Flowchart

DETERMINATION OF THE CALIBRATION PARAMETERS

Analog calibration algorithm sampling requirements

For optimal calibration results at least 128 samples per each magnetic period (N+S magnetic pole pair, also referred to as 360° electrical period) should be recorded. This corresponds to one sample at least every 2.8° electrical. Decreasing this distance to less than 2.8° is beneficial to the calibration algorithm performance. The quality of the calibration results slowly deteriorates with sample distance greater than 2.8° electrical. Recording multiple rotations or linear movements and a high sampling rate significantly increases the probability to record a sample point at least every 2.8° electrical.

Please see Figure 24 below for examples of good and poor data acquisition results.

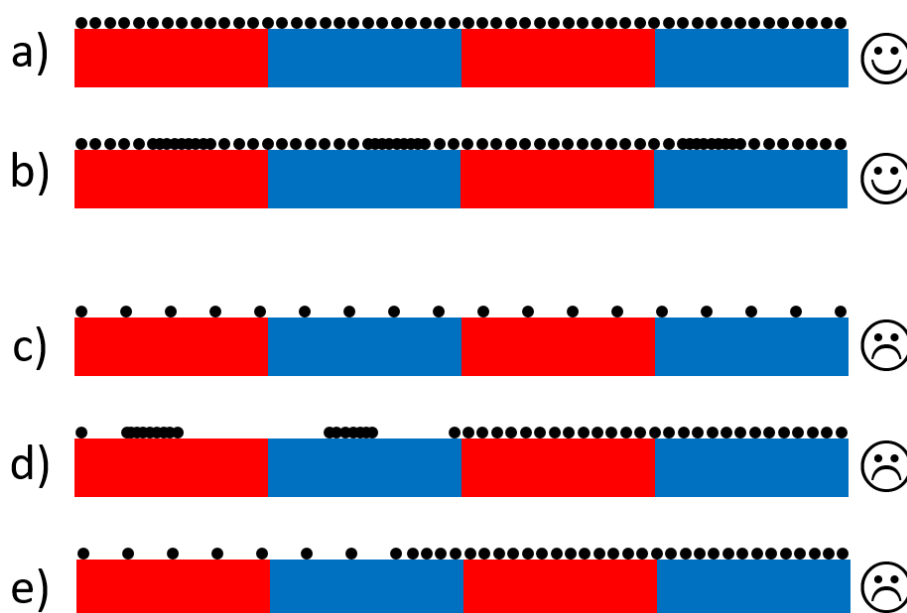


Figure 24:

- a) and b): good data acquisition results
- c) sample distance greater than 2.8° electrical
- d) irregular sampling with gaps greater than 2.8° electrical
- e) magnetic period(s) with sample distance greater than 2.8° electrical

Calibration via BiSS interface

A BiSS frame rate of 31.25µs is practical in most application scenarios and allows reasonable rotation or linear speeds during calibration data acquisition. Faster BiSS frame rates may be chosen if possible. For lower BiSS clock frequencies or low performance PCs it might be necessary to slow down the BiSS frame rate e.g. to 62.5µs or even slower to avoid data loss during calibration data acquisition.

Rotary Analog Calibration

Due to the requirements of at least one sample per 2.8° electrical (128 samples per magnetic period) the maximum recommended rotational speed for a specific BiSS Frame Rate can be calculated as:

$$RPM_{max} \left[\frac{1}{min} \right] = \frac{60}{\text{BiSS Frame Rate [s]} \times \text{Number Of Magnetic Master Periods} \times 128}$$

Example: 31.25 µs BiSS Frame Rate and 32 master pole pair magnetic pole wheel:

$$\frac{60}{31.25 \times 10^{-6} \text{ s} \times 32 \times 128} = 468.8 \left[\frac{1}{\text{min}} \right]$$

Linear Analog Calibration

The maximum recommended linear speed for a specific BiSS Frame Rate can be calculated as:

$$V_{\text{max}} \left[\frac{\text{m}}{\text{s}} \right] = \frac{\text{Magnetic Period Length [m]}}{\text{BiSS Frame Rate [s]} \times 128}$$

Example: 31.25 µs BiSS Frame Rate and 4 mm magnetic period:

$$\frac{0.004 \text{ m}}{31.25 \times 10^{-6} \text{ s} \times 128} = 1 \left[\frac{\text{m}}{\text{s}} \right]$$

Calibration via MB3U-I2C SPI interface

For the MB3U-I2C adapter the SPI Frame Rate can be estimated as follows:

$$\text{SPIFrameRate [s]} = \frac{40}{\text{SPI Frequency [Hz]}} + 3 \times 10^{-6} \text{ [s]}$$



If the data acquisition is done with own SPI hardware, the frame rate has to be determined individually.

Rotary Analog Calibration

Due to the requirements of at least one sample per 2.8° electrical (128 samples per magnetic period) the maximum recommended rotation speed for a specific SPI frequency can be calculated as:

$$\text{RPM}_{\text{max}} \left[\frac{1}{\text{min}} \right] = \frac{60}{\text{SPI Frame Rate [s]} \times \text{Number Of Magnetic Master Periods} \times 128}$$

Example: The maximum permissible rotational speed for 100 kHz SPI frequency and 32 master pole pair magnetic pole wheel can be calculated as:

$$\frac{60}{403 \times 10^{-6} \text{ s} \times 32 \times 128} = 36.3 \left[\frac{1}{\text{min}} \right]$$

Linear Analog Calibration

The maximum recommended linear speed for a specific SPI frequency can be calculated as:

$$V_{\text{max}} \left[\frac{\text{m}}{\text{s}} \right] = \frac{\text{Magnetic Period Length [m]}}{\text{SPI Frame Rate [s]} \times 128}$$

Example: The maximum permissible linear speed for 100 kHz SPI frequency and 4 mm magnetic period can be calculated as:

$$\frac{0.004 \text{ m}}{403 \times 10^{-6} \text{ s} \times 128} = 0.0775 \left[\frac{\text{m}}{\text{s}} \right]$$



A "Calibration Quality Check" feature is integrated in the iC-MU Series GUI (also see page 7). It can be used to automatically calculate the recommended rotary and linear calibration speeds based on the formulas described above and actual GUI data acquisition and chip settings.

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ROTARY CALIBRATION AND PROGRAMMING



Rev E1, Page 21/21

REVISION HISTORY

Rel.	Rel. Date*	Chapter	Modification	Page
D1	2022-02-02	All	Initial release for GUI revision B1 or later.	all

Rel.	Rel. Date*	Chapter	Modification	Page
E1	2023-08-28	All	Images throughout the document updated according to the latest GUI revision. Updated image descriptions.	all
		TOOLS DOCUMENTATION FOR CALIBRATION AND PROGRAMMING	Link section cleaned, Added Table 2: "USB PC adapter compatibility matrix for iC-MU Series calibration".	1
		CALIBRATION	Reworked section "Calibration Data Acquisition Settings", Added section "Calibration Quality Check Settings"	7
		CALIBRATION	Added advice to deactivate the cyclic multiturn verification during calibration, added info that also only a section of a pole wheel can be scanned for calibration.	10
		CALIBRATION	Added typical Nonius track offset values.	11
		DETERMINATION OF THE CALIBRATION PARAMETERS	Renamed chapter "ANNEX" into "DETERMINATION OF THE CALIBRATION PARAMETERS", reworked chapter to reflect the requirements of the latest calibration algorithm.	18 - 20

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* Release Date format: YYYY-MM-DD